

Conversion from Mealy machine to Moore Machine

In Moore machine, the output is associated with every state, and in Mealy machine, the output is given along the edge with input symbol. To convert Moore machine to Mealy machine, state output symbols are distributed to input symbol paths. But while converting the Mealy machine to Moore machine, we will create a separate state for every new output symbol and according to incoming and outgoing edges are distributed.

The following steps are used for converting Mealy machine to the Moore machine:

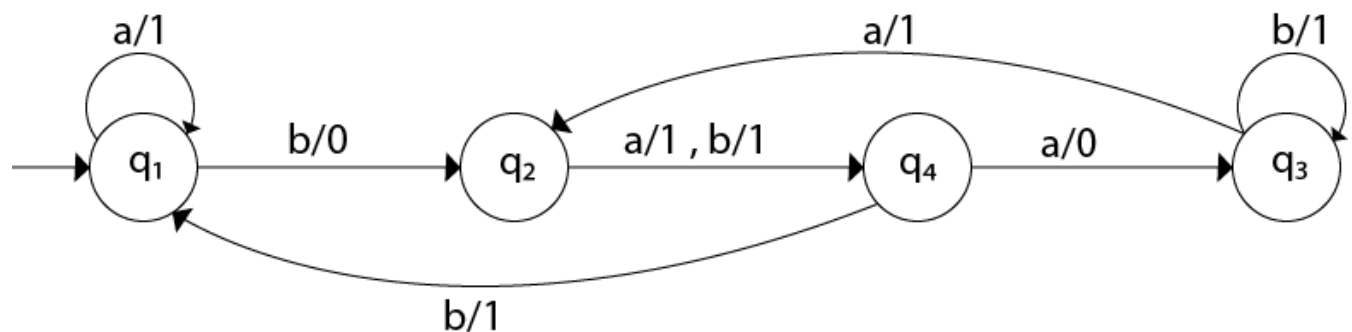
Step 1: For each state (Q_i), calculate the number of different outputs that are available in the transition table of the Mealy machine.

Step 2: Copy state Q_i , if all the outputs of Q_i are the same. Break q_i into n states as Q_{in} , if it has n distinct outputs where $n = 0, 1, 2, \dots$

Step 3: If the output of initial state is 0, insert a new initial state at the starting which gives 1 output.

Example 1:

Convert the following Mealy machine into equivalent Moore machine.



Solution:

Transition table for above Mealy machine is as follows:

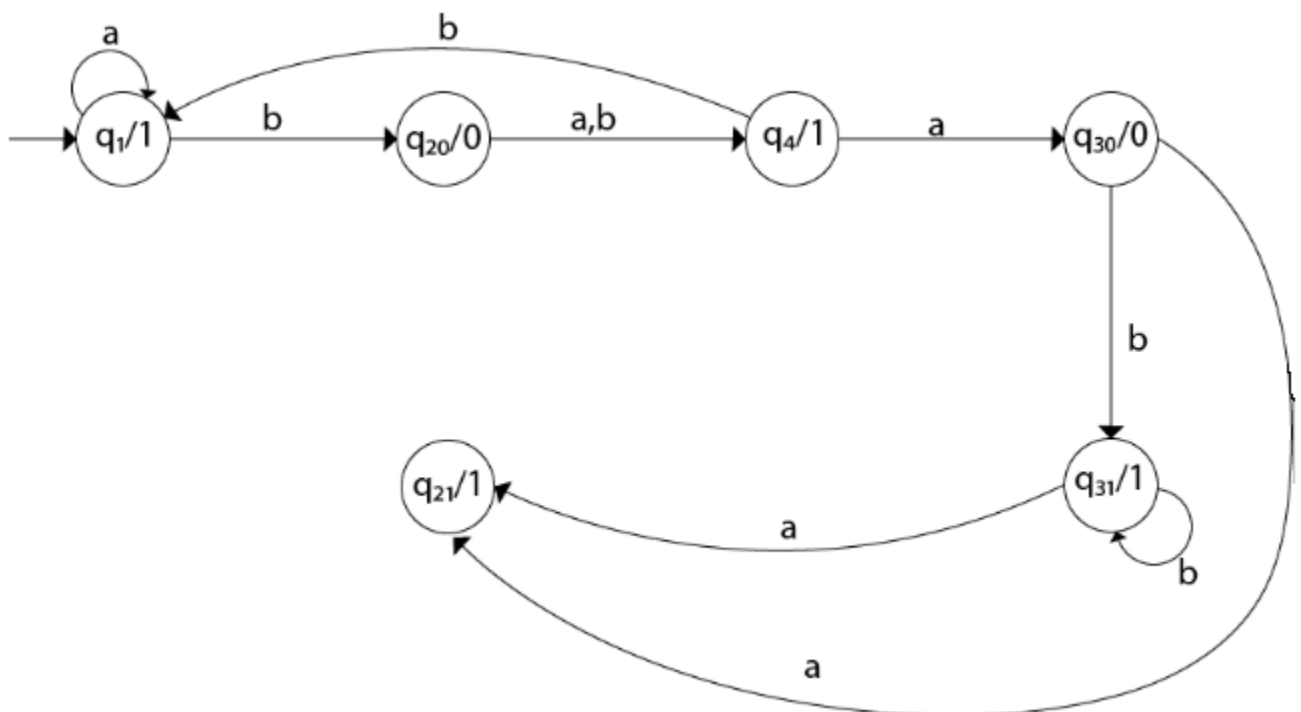
Present State	Next State			
	a		b	
	State	O/P	State	O/P
q ₁	q ₁	1	q ₂	0
q ₂	q ₄	1	q ₄	1
q ₃	q ₂	1	q ₃	1
q ₄	q ₃	0	q ₁	1

- For state q₁, there is only one incident edge with output 0. So, we don't need to split this state in Moore machine.
- For state q₂, there is 2 incident edge with output 0 and 1. So, we will split this state into two states q₂₀(state with output 0) and q₂₁(with output 1).
- For state q₃, there is 2 incident edge with output 0 and 1. So, we will split this state into two states q₃₀(state with output 0) and q₃₁(state with output 1).
- For state q₄, there is only one incident edge with output 0. So, we don't need to split this state in Moore machine.

Transition table for Moore machine will be:

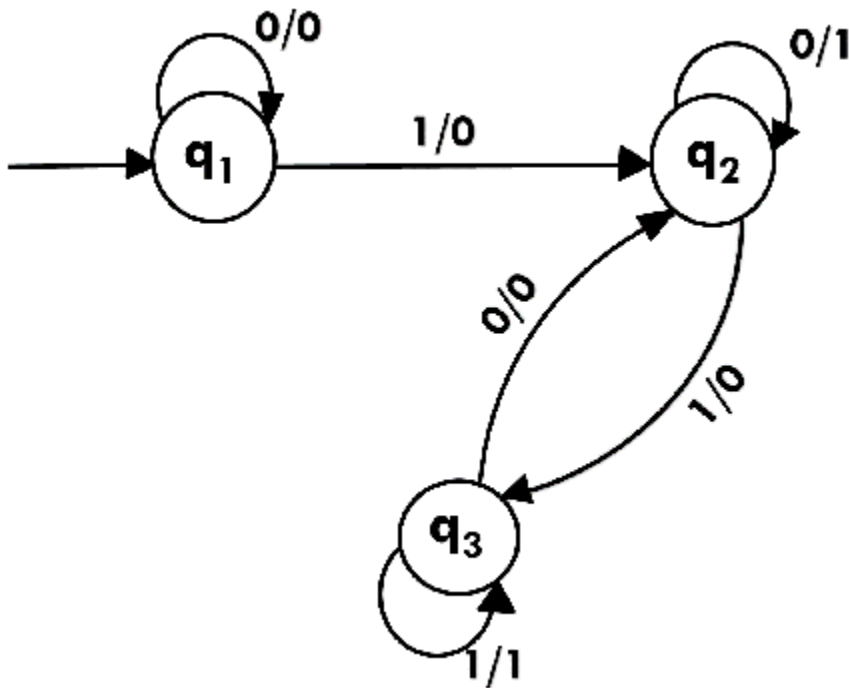
Present State	Next State		Output
	a=0	a=1	
q ₁	q ₁	q ₂	1
q ₂₀	q ₄	q ₄	0
q ₂₁	∅	∅	1
q ₃₀	q ₂₁	q ₃₁	0
q ₃₁	q ₂₁	q ₃₁	1
q ₄	q ₃	q ₄	1

Transition diagram for Moore machine will be:



Example 2:

Convert the following Mealy machine into equivalent Moore machine.



Solution:

Transition table for above Mealy machine is as follows:

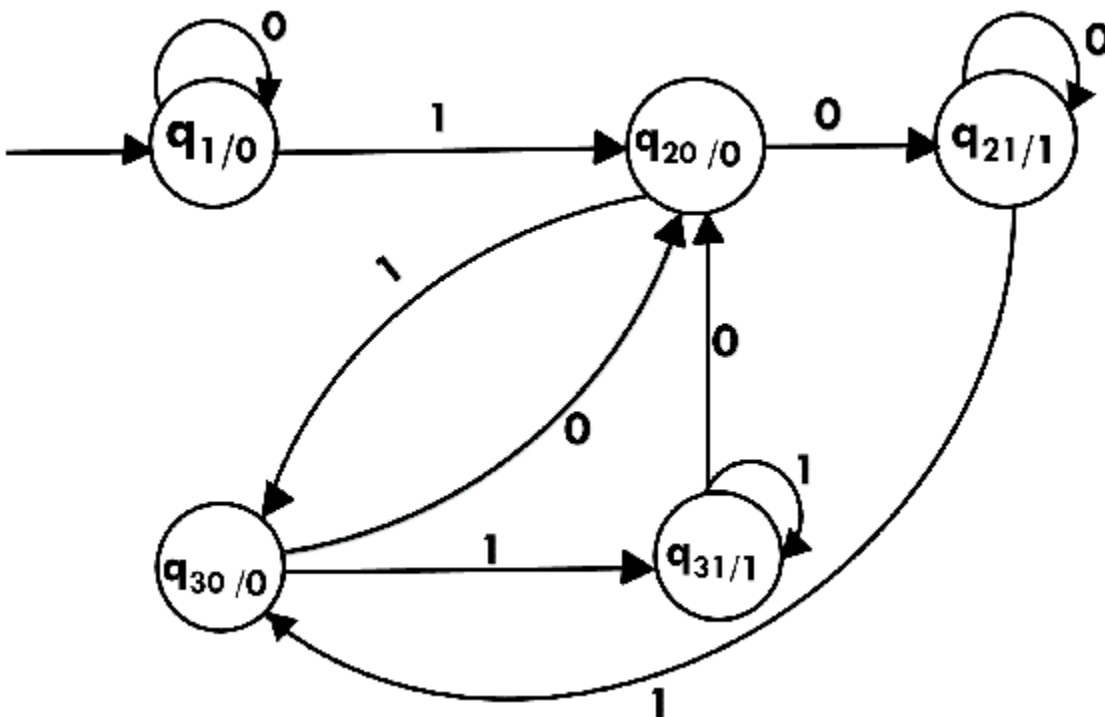
Present State	Next State 0		Next State 1	
	State	o/P	State	o/P
q₁	q₁	0	q₂	0
q₂	q₂	1	q₃	0
q₃	q₂	0	q₃	1

The state q_1 has only one output. The state q_2 and q_3 have both output 0 and 1. So we will create two states for these states. For q_2 , two states will be q_{20} (with output 0) and q_{21} (with output 1). Similarly, for q_3 two states will be q_{30} (with output 0) and q_{31} (with output 1).

Transition table for Moore machine will be:

Present State	Next State 0	Next State 1	o/P
q_1	q_1	q_{20}	0
q_{20}	q_{21}	q_{30}	0
q_{21}	q_{21}	q_{30}	1
q_{30}	q_{20}	q_{31}	0
q_{31}	q_{20}	q_{31}	1

Transition diagram for Moore machine will be:



Conversion from Moore machine to Mealy Machine

In the Moore machine, the output is associated with every state, and in the mealy machine, the output is given along the edge with input symbol. The equivalence of the Moore machine and Mealy machine means both the machines generate the same output string for same input string.

We cannot directly convert Moore machine to its equivalent Mealy machine because the length of the Moore machine is one longer than the Mealy machine for the given input. To convert Moore machine to Mealy machine, state output symbols are distributed into input symbol paths. We are going to use the following method to convert the Moore machine to Mealy machine.

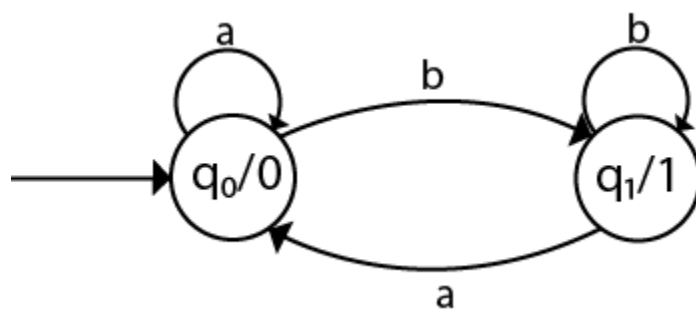
Method for conversion of Moore machine to Mealy machine

Let $M = (Q, \Sigma, \delta, \lambda, q_0)$ be a Moore machine. The equivalent Mealy machine can be represented by $M' = (Q, \Sigma, \delta, \lambda', q_0)$. The output function λ' can be obtained as:

$$\lambda'(q, a) = \lambda(\delta(q, a))$$

Example 1:

Convert the following Moore machine into its equivalent Mealy machine.



Solution:

The transition table of given Moore machine is as follows:

Q	a	b	Output(λ)
q0	q0	q1	0
q1	q0	q1	1

The equivalent Mealy machine can be obtained as follows:

$$\begin{aligned}
 \lambda'(q_0, a) &= \lambda(\delta(q_0, a)) \\
 &= \lambda(q_0) \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 \lambda'(q_0, b) &= \lambda(\delta(q_0, b)) \\
 &= \lambda(q_1) \\
 &= 1
 \end{aligned}$$

The λ for state q1 is as follows:

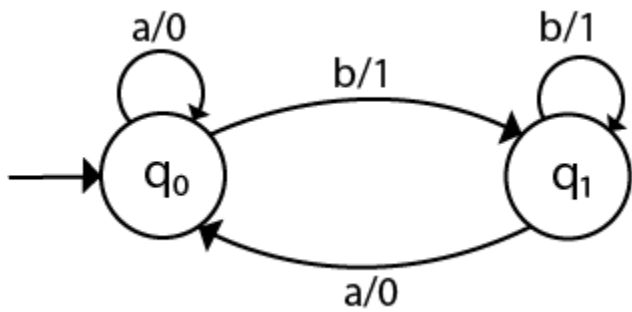
$$\begin{aligned}
 \lambda'(q_1, a) &= \lambda(\delta(q_1, a)) \\
 &= \lambda(q_0) \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 \lambda'(q_1, b) &= \lambda(\delta(q_1, b)) \\
 &= \lambda(q_1) \\
 &= 1
 \end{aligned}$$

Hence the transition table for the Mealy machine can be drawn as follows:

Q \ Σ	Input 0		Input 1	
	State	O/P	State	O/P
q ₀	q ₀	0	q ₁	1
q ₁	q ₀	0	q ₁	1

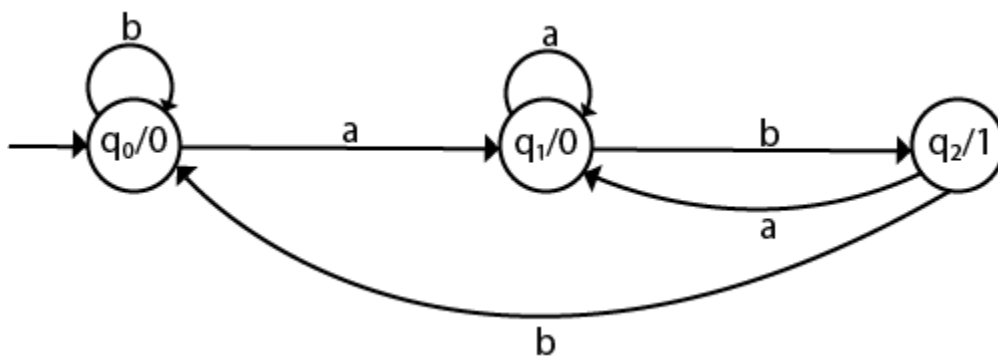
The equivalent Mealy machine will be,



Note: The length of output sequence is ' $n+1$ ' in Moore machine and is ' n ' in the Mealy machine.

Example 2:

Convert the given Moore machine into its equivalent Mealy machine.



Solution:

The transition table of given Moore machine is as follows:

Q	a	b	Output(λ)
q0	q1	q0	0
q1	q1	q2	0

q2	q1	q0	1
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The equivalent Mealy machine can be obtained as follows:

$$\begin{aligned}
 \lambda' (q_0, a) &= \lambda(\delta(q_0, a)) \\
 &= \lambda(q_1) \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 \lambda' (q_0, b) &= \lambda(\delta(q_0, b)) \\
 &= \lambda(q_0) \\
 &= 0
 \end{aligned}$$

The λ for state q1 is as follows:

$$\begin{aligned}
 \lambda' (q_1, a) &= \lambda(\delta(q_1, a)) \\
 &= \lambda(q_1) \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 \lambda' (q_1, b) &= \lambda(\delta(q_1, b)) \\
 &= \lambda(q_2) \\
 &= 1
 \end{aligned}$$

The λ for state q2 is as follows:

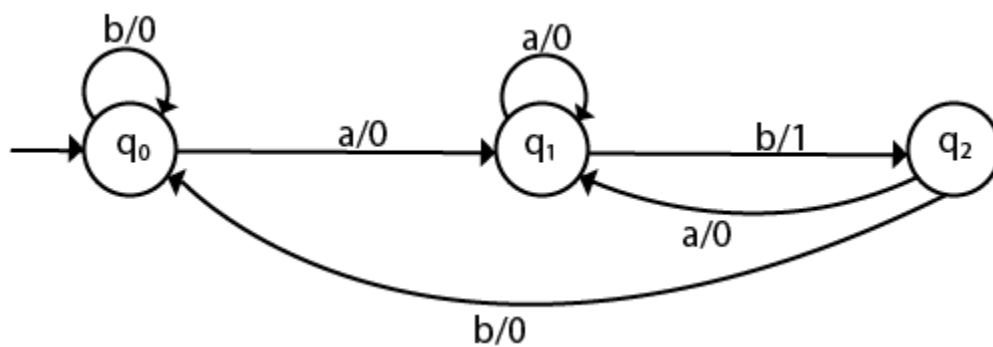
$$\begin{aligned}
 \lambda' (q_2, a) &= \lambda(\delta(q_2, a)) \\
 &= \lambda(q_1) \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 \lambda' (q_2, b) &= \lambda(\delta(q_2, b)) \\
 &= \lambda(q_0) \\
 &= 0
 \end{aligned}$$

Hence the transition table for the Mealy machine can be drawn as follows:

Σ Q	Input a		Input b	
	State	Output	State	Output
q ₀	q ₁	0	q ₀	0
q ₁	q ₁	0	q ₂	1
q ₂	q ₁	0	q ₀	0

The equivalent Mealy machine will be,



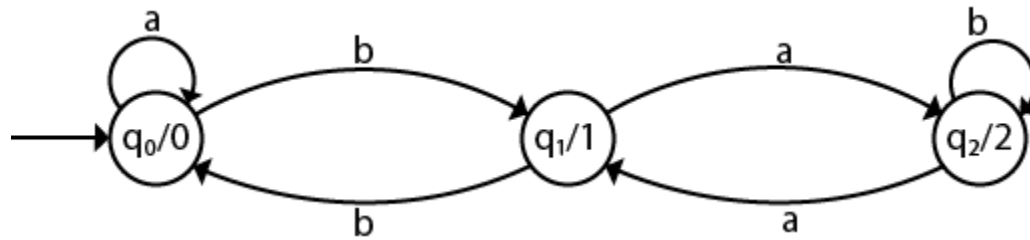
Example 3:

Convert the given Moore machine into its equivalent Mealy machine.

Q	a	b	Output(λ)
q ₀	q ₀	q ₁	0
q ₁	q ₂	q ₀	1
q ₂	q ₁	q ₂	2

Solution:

The transaction diagram for the given problem can be drawn as:



The equivalent Mealy machine can be obtained as follows:

$$\begin{aligned}\lambda'(q_0, a) &= \lambda(\delta(q_0, a)) \\ &= \lambda(q_0) \\ &= 0\end{aligned}$$

$$\begin{aligned}\lambda'(q_0, b) &= \lambda(\delta(q_0, b)) \\ &= \lambda(q_1) \\ &= 1\end{aligned}$$

The λ for state q_1 is as follows:

$$\begin{aligned}\lambda'(q_1, a) &= \lambda(\delta(q_1, a)) \\ &= \lambda(q_2) \\ &= 2\end{aligned}$$

$$\begin{aligned}\lambda'(q_1, b) &= \lambda(\delta(q_1, b)) \\ &= \lambda(q_0) \\ &= 0\end{aligned}$$

The λ for state q_2 is as follows:

$$\begin{aligned}\lambda'(q_2, a) &= \lambda(\delta(q_2, a)) \\ &= \lambda(q_1) \\ &= 1\end{aligned}$$

$$\begin{aligned}
 \lambda'(q_2, b) &= \lambda(\delta(q_2, b)) \\
 &= \lambda(q_2) \\
 &= 2
 \end{aligned}$$

Hence the transition table for the Mealy machine can be drawn as follows:

Q \ Σ	Input a		Input b	
	State	O/P	State	O/P
q_0	q_0	0	q_1	1
q_1	q_2	2	q_0	0
q_2	q_1	1	q_2	2

The equivalent Mealy machine will be,

